Move

Global Decentralized Transport Discovery
and Booking Infrastructure on BOT Chain

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A permissionless platform where anyone, anywhere can search routes,
book seats and settle payments with BOT or USDT — every record lives on-chain.

Built on BOT Chain · Chain ID 677

https://botchain.ai · https://scan.botchain.ai

1. Abstract

Move is a fully decentralized transport discovery and booking platform deployed on BOT Chain. It enables any passenger worldwide to search routes, lock seats and pay with native BOT or USDT, while any operator can register vehicles, publish schedules and receive instant on-chain settlements. All permanent state — operators, vehicles, routes, bookings, payments, cancellations and reviews — is stored immutably on BOT Chain. There is no traditional backend or database. This whitepaper describes the problem, architecture, workflows, payment design, security model and roadmap.

2. Introduction

Intercity and inter-regional transport remains highly fragmented across most markets. Passengers face opaque pricing, uncertain seat availability and cash-heavy settlement. Operators struggle with manual inventory management and delayed reconciliation. Existing digital platforms rely on centralized servers that introduce single points of failure and limited transparency. Move removes the intermediary layer by placing the entire booking lifecycle on BOT Chain — an EVM-compatible Layer-1 with ~0.75-second blocks and extremely low fees, ideal for frequent, small-value mobility transactions.

3. Problem Statement

There is no widely adopted, permissionless, fully on-chain infrastructure that lets any passenger anywhere discover verified routes, lock seats and pay digitally while letting any operator publish schedules and receive trustless settlements — without a centralized backend.

4. Platform Vision

Move aims to become the open transport booking layer for the BOT Chain ecosystem and beyond. Any origin, any destination. Any operator. Any passenger. Every booking, payment and review becomes a permanent public record. The system is intentionally backend-free: smart contracts plus a static frontend constitute the entire application.

5. Proposed Solution

A single primary smart contract (Move.sol) on BOT Chain together with a vanilla HTML/CSS/JS frontend using ethers.js. The contract manages operator registration & verification, vehicle registration, route & schedule creation, booking with BOT or USDT, automatic fee split and operator payout, cancellation windows with percentage refunds, and on-chain reviews. Seat availability is decremented atomically at booking time, preventing double-booking.

6. Passenger Workflow

1. Connect a BOT Chain wallet (MetaMask, BO Wallet, etc.). Network is validated automatically.
2. Search any origin and destination. Compare fares, seats, times and operators.
3. Select seats, enter details, choose BOT or USDT, and confirm the transaction.
4. View active bookings, generate a booking QR, cancel eligible trips, or leave reviews.

7. Operator Workflow

1. Connect wallet and register a company profile.
2. Optionally await administrator verification for full publishing rights.
3. Register vehicles (plate, type, model, capacity).
4. Publish routes with origin, destination, departure time, duration and fares.

5. Receive net proceeds automatically; monitor bookings and handle cancellations.

8. Route Discovery

Routes are published exclusively by verified operators. The frontend calls an on-chain search that filters active routes matching origin and destination with future departures and remaining seats. No fabricated availability is shown; seat counts always reflect contract state. For production scale, event indexing is recommended while the contract remains the source of truth.

9. Payment Architecture

Native BOT — Passenger sends BOT with the booking transaction. The contract deducts the platform fee (configurable basis points) and transfers the remainder to the operator wallet.

USDT — Passenger approves the Move contract to spend the required USDT amount (BOT Chain USDT at 0xaBabc7Ddc03e501d190C676BF3d92ef0e6e87a3C, 6 decimals). The booking function pulls the funds, splits the fee and pays the operator. All transfers are visible on the explorer.

10. Booking, Cancellation & Refunds

Seat availability is checked and decremented inside the same transaction as payment. A booking may be cancelled by the passenger, operator or platform owner if the current time is at least two hours before departure. Upon cancellation seats are restored and 90% of the paid amount is refunded automatically; 10% is retained as a cancellation fee. Rules are enforced on-chain.

11. Security

- Value transfers occur only inside booking and cancellation functions.
- Platform fee is hard-capped at 10% in the setter.
- Only the contract owner can verify operators, update fees or withdraw accumulated fees.
- USDT transfers use standard ERC-20 patterns with explicit success checks.
- Arithmetic is overflow-checked (Solidity 0.8+).

12. Roadmap

Phase 1 — Core contract, passenger & operator dashboards, BOT/USDT payments, search, whitepaper.

Phase 2 — Event indexing, PWA, enhanced QR boarding, operator verification workflow.

Phase 3 — Multi-leg itineraries, reputation scoring, agent integrations, fee governance.

Phase 4 — Broader geographic coverage, formal audits, insurance primitives.

13. Conclusion

Move proves that a complete, global mobility marketplace can operate without a traditional backend by placing every permanent record on BOT Chain. Fast finality, low fees, native BOT and bridged USDT make the network an ideal settlement layer for consumer transport. The platform is open for inspection, deployment and community-driven improvement.

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